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United States Patent Application Publication

20250286611

Kind Code

A1

Publication Date

September 11, 2025

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MULTI-BAND RADIOFREQUENCY TRANSCEIVER

Abstract

In some implementations, a satellite gateway includes a set of modulators comprising (i) a first modulator to modulate signals for transmission using a first of multiple polarizations, and (ii) a second modulator to modulate signals for transmission using a second of the multiple polarizations. The satellite gateway includes a set of upconverters comprising at least one upconverter for each of the multiple polarizations, wherein each of the upconverters is configured to generate a radiofrequency output by concurrently upconverting multiple intermediate frequency inputs to different frequency ranges. The satellite gateway includes an intermediate frequency distribution network configured to distribute the multiple intermediate frequency output signals from each of the modulators to the inputs of the upconverters. The satellite gateway includes a set of radiofrequency switches that are operable to selectively provide the radiofrequency outputs of the upconverters to antenna feeds for the multiple polarizations.

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Family ID: 1000008618297

Appl. No.: 19/213639

Filed: May 20, 2025

Related U.S. Application Data

parent US division 17941939 20220909 parent-grant-document US 12335025 child US 19213639
us-provisional-application US 63242466 20210909

Publication Classification

Int. Cl.: H04B7/185 (20060101); **H04B1/00** (20060101); **H04B1/04** (20060101); **H04B1/16** (20060101)

U.S. Cl.:

CPC H04B7/18517 (20130101); **H04B1/0064** (20130101); **H04B1/0075** (20130101); **H04B1/04** (20130101); **H04B1/16** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 17/941,939, filed on Sep. 9, 2022, which claims the benefit of priority to U.S. Provisional Patent Application No. 63/242,466, filed on Sep. 9, 2021, the entire contents of which are incorporated by reference herein.

FIELD

[0002] This specification generally describes technology related to providing improved receivers and transmitters for communication systems, such as for gateways of satellite networks.

BACKGROUND

[0003] In a satellite communication system, a gateway is used as a ground station to connect a terrestrial network, such as the Internet, and satellites orbiting Earth. For example, the gateway can transmit data from user terminals or other devices to the orbiting satellites. Further, the gateway can receive data from the orbiting satellites and relay the data to the terrestrial network. The gateway uses antennas and other equipment for converting, transmitting, and receiving signals. For example, the gateway typically includes a down converter and demodulator to receive signals from the satellite. Similarly, the gateway typically includes a modulator, an upconverter, and a power amplifier to transmit signals to the satellite to be related to satellite terminals.

SUMMARY

[0004] In some implementations, a satellite gateway includes a receiver subsystem and a transmitter subsystem that are each configured to operate using multiple polarizations and multiple frequency channel segments. The receiver subsystem includes downconverters that each provide output for multiple bands. For example, each downconverter can provide multiple intermediate frequency outputs derived from the same radiofrequency input. Similarly, the transmitter subsystem includes upconverters that each upconvert multiple intermediate frequency input signals to different frequency ranges. Each upconverter provides a radiofrequency output that combines the modulated signals from multiple channels, each upconverted to the desired frequency range for the channel.

[0005] The satellite gateway can be configured to transmit and receive information for multiple different polarizations, such as right-hand circular polarization (RHCP) and left-hand circular polarization (LHCP). To provide resilience in the event of potential component failures, the receiver subsystem and/or the transmitter subsystem may use an intermediate frequency (IF) distribution network that enables the signals for different polarizations to be routed to multiple destinations (e.g., different demodulators for reception, or different upconverters for transmission). For example, the IF distribution network may include a switch matrix providing programmable routing of IF signals. As another example, the IF distribution network may include radiofrequency switches that enable cross-switching between different signal paths. This can enable dynamic selection of a straight coupling (e.g., signals for RHCP transmission provided to an upconverter for RHCP transmission) or cross-coupling (e.g., signals for RHCP transmission provided to a upconverter for LHCP transmission). Other IF distribution network options can include using

signal dividers or splitters to provide IF signals to multiple components, or using summing elements to combine signals across inputs from different IF signals sources.

[0006] In one general aspect, a transmitter system includes: a set of modulators comprising (i) one or more first modulators to modulate signals for transmission using a first of multiple polarizations, and (ii) one or more second modulators to modulate signals for transmission using a second of the multiple polarizations, wherein the one or more modulators for each polarization are configured to provide multiple intermediate frequency outputs that overlap in frequency range and carry different data; a set of upconverters comprising at least one upconverter for each of the multiple polarizations, wherein each of the upconverters is configured to generate a radiofrequency output by concurrently upconverting multiple intermediate frequency inputs to different frequency ranges; an intermediate frequency distribution network configured to distribute the multiple intermediate frequency output signals from each of the modulators to the inputs of the upconverters; and a set of radiofrequency switches that are operable to selectively provide the radiofrequency outputs of the upconverters to antenna feeds for the multiple polarizations.

[0007] Implementations can include one or more of the following features. For example, in some implementations, each of the upconverters is configured to receive and upconvert three or four intermediate frequency signals to generate the radiofrequency output of the upconverter.

[0008] In some implementations, the intermediate frequency distribution network comprises a switch matrix that is programmable to selectively provide intermediate frequency outputs from any of the modulators to any of the upconverters.

[0009] In some implementations, the intermediate frequency distribution network comprises a set of switches that are programmable to selectively provide cross-switching to change routing of the intermediate frequency outputs from the modulators to different upconverters. In some implementations, the modulators each have a primary output and a secondary output for each intermediate frequency output, and each of the modulators includes a switch to select between providing output on the primary output or the secondary output; and the intermediate frequency distribution network comprises a set of summing elements that each combine (1) intermediate frequency signals on a primary output of a modulator for the first polarization and (2) intermediate frequency signals on a secondary output of a modulator for the first polarization, the intermediate frequency distribution network being configured to provide the combined output of the respective summing elements as inputs to the upconverters.

[0010] In some implementations, the multiple polarizations include right-hand circular polarization and left-hand circular polarization.

[0011] In some implementations, the transmitter system is configured to transmit signals in the V band, over a bandwidth of at least 4 GHz.

[0012] In some implementations, the transmitter system is configured to transmit data for a plurality of different channel segments, and each of the multiple intermediate frequency outputs of the modulators includes modulated signals for multiple of the channel segments.

[0013] In some implementations, the transmitter system is configured to transmit signals for each of the multiple polarizations in multiple channel segments each having bandwidth between 250 MHz and 2 GHz.

[0014] In some implementations, the transmitter system is configured to transmit signals for each of the multiple polarizations in at least eight channel segments each having bandwidth of substantially 500 MHz.

[0015] In another general aspect, a receiver system includes: a set of downconverters comprising (i) a first downconverter to downconvert signals received for a first of multiple polarizations, and (ii) a second downconverter to downconvert signals received for a second of the multiple polarizations, wherein each of the downconverters is configured to provide multiple intermediate frequency outputs that overlap in frequency range and carry different data; a set of demodulators comprising at least one demodulator for each of the multiple polarizations, wherein each of the

demodulators is configured to extract data from a different intermediate frequency signal; an intermediate frequency distribution network configured to distribute the multiple intermediate frequency output signals from each of the downconverters to the inputs of the demodulators; and a network interface configured to provide extracted data to a communication network.

[0016] Implementations can include one or more of the following features. For example, each of the downconverters is configured to output three or four intermediate frequency signals based on an input radiofrequency signal.

[0017] In some implementations, the receiver system includes a third downconverter; and a set of radiofrequency switches that are operable to (i) selectively switch radiofrequency input for the first of the multiple polarizations from the first downconverter to the third downconverter, and (i) selectively switch radiofrequency input for the second of the multiple polarizations from the second downconverter to the third downconverter.

[0018] In some implementations, the intermediate frequency distribution network comprises a switch matrix that is programmable to selectively provide intermediate frequency outputs from any of the downconverters to at least one demodulator in any of multiple different groups of demodulators.

[0019] In some implementations, the intermediate frequency distribution network comprises frequency dividers for each down converter, the frequency dividers providing the output of a downconverter to each of multiple sets of demodulators.

[0020] In some implementations, the multiple polarizations include right-hand circular polarization and left-hand circular polarization.

[0021] In some implementations, the receiver system is configured to receive signals in the Q band, over a bandwidth of at least 4 GHz.

[0022] In some implementations, the receiver system is configured to receive data for a plurality of different channel segments, and each of the multiple intermediate frequency outputs of the downconverters includes modulated signals for multiple of the channel segments.

[0023] In some implementations, the receiver system is configured to receive signals for each of the multiple polarizations in multiple channel segments each having bandwidth between 250 MHz and 2 GHz.

[0024] In some implementations, the receiver system is configured to receive signals for each of the multiple polarizations in at least eight channel segments each having bandwidth of substantially 500 MHz.

[0025] In another general aspect, a satellite gateway includes a receiver subsystem and a transmitter subsystem. The transmitter subsystem includes: a set of modulators comprising (i) one or more first modulators to modulate signals for transmission using a first of multiple polarizations, and (ii) one or more second modulators to modulate signals for transmission using a second of the multiple polarizations, wherein the one or more modulators for each polarization are configured to provide multiple intermediate frequency outputs that overlap in frequency range and carry different data; a set of upconverters comprising at least one upconverter for each of the multiple polarizations, wherein each of the upconverters is configured to generate a radiofrequency output by concurrently upconverting multiple intermediate frequency inputs to different frequency ranges; an intermediate frequency distribution network configured to distribute the multiple intermediate frequency output signals from each of the modulators to the inputs of the upconverters; and a set of radiofrequency switches that are operable to selectively provide the radiofrequency outputs of the upconverters to antenna feeds for the multiple polarizations. The receiver subsystem includes: a set of downconverters comprising (i) a first downconverter to downconvert signals received for a first of multiple polarizations, and (ii) a second downconverter to downconvert signals received for a second of the multiple polarizations, wherein each of the downconverters is configured to provide multiple intermediate frequency outputs that overlap in frequency range and carry different data; a set of demodulators comprising at least one demodulator for each of the multiple polarizations,

wherein each of the demodulators is configured to extract data from a different intermediate frequency signal; an intermediate frequency distribution network configured to distribute the multiple intermediate frequency output signals from each of the downconverters to the inputs of the demodulators; and a network interface configured to provide extracted data to a communication network.

[0026] In some implementations, the satellite gateway is configured to receive signals in the Q band, and to transmit signals in the V band.

[0027] In some implementations, the satellite gateway is configured to receive over a bandwidth of at least 4.5 GHz.

[0028] In some implementations, the satellite gateway is configured to transmit over at bandwidth of at least 4 GHz.

[0029] In some implementations, the satellite gateway is configured to receive signals over a range of 37.5-42.5 GHz.

[0030] In some implementations, the satellite gateway is configured to transmit signals over ranges of 47.2-50.2 GHz and 50.4-51.4 GHz.

[0031] In some implementations, the satellite gateway is configured to receive signals 9 or 10 channel segments of substantially 500 MHz.

[0032] In some implementations, the satellite gateway is configured to transmit signals for at least 8 channel segments of substantially 500 MHz.

[0033] In some implementations, the satellite gateway and an associated antenna are configured to achieve cross-polar discrimination (XPD) of at least 30 dB.

[0034] In some implementations, the satellite gateway is configured to use intermediate frequencies in range from 0.5 to 5.0 GHz.

[0035] In some implementations, the intermediate frequency signals includes L band signals and/or S band signals.

[0036] In another general aspect, a method includes: receiving, over a network interface, data for transmission in a satellite communication system; generating multiple intermediate frequency outputs for each of multiple polarizations by modulating one or more carriers based on the data for transmission using one or more first modulators for a first polarization and one or more modulators for a second polarization, wherein the one or more modulators for each polarization are configured to provide multiple intermediate frequency outputs that overlap in frequency range and carry different data; distributing the multiple intermediate frequency output signals for each polarization from the modulators to the inputs of a set of upconverters, the set of upconverters comprising at least one upconverter for each of the multiple polarizations; upconverting, for each polarization, the multiple intermediate frequency outputs for the polarization using the at least one upconverter for the polarization, wherein each of the upconverters is configured to generate a radiofrequency output by concurrently upconverting multiple intermediate frequency inputs to different frequency ranges; and routing the radiofrequency outputs to antenna feeds using a set of radiofrequency switches.

[0037] In another general aspect, a method comprises: receiving radiofrequency signals for each of the multiple polarizations; downconverting the radiofrequency signals using (i) a first downconverter to downconvert signals received for the first polarization, and (ii) a second downconverter to downconvert signals received for the second polarization, wherein each of the downconverters is configured to provide multiple intermediate frequency outputs that overlap in frequency range and carry different data; distributing the multiple intermediate frequency output signals from the downconverters to inputs of a set of demodulators using a second intermediate frequency distribution network, the set of demodulators comprising at least one demodulator for each of the multiple polarizations; demodulating the multiple intermediate frequency output signals from the downconverters using the set of demodulators; and providing, to a communication network, extracted data obtained from demodulating the multiple intermediate frequency output

signals.

[0038] The details of one or more implementations are set forth in the accompanying drawings and the description below. Other potential features and advantages will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0039] FIG. 1 is a diagram that illustrates an example of a satellite networking system.

[0040] FIG. 2 is a diagram that illustrates example frequency plans for uplink and downlink for a satellite gateway of the satellite networking system.

[0041] FIGS. 3A-3B are diagrams that illustrate various uplink frequency plans for the satellite gateway.

[0042] FIGS. 4A-4F are diagrams that illustrate various example transmitter subsystems for the satellite gateway.

[0043] FIGS. 5A-5D are diagrams that illustrate various downlink frequency plans for the satellite gateway.

[0044] FIGS. 6A-6G are diagrams that illustrate various example transmitter subsystems for the satellite gateway.

[0045] FIG. 7 is a diagram that illustrates parameters and characteristics of different channel and intermediate frequency options for downlink communication in the satellite gateway.

[0046] In the drawings, like reference numbers represent corresponding parts throughout.

DETAILED DESCRIPTION

[0047] FIG. 1 shows an example of a satellite communication system **100**. The system **100** includes a satellite gateway **110** that communicates with one or more satellites **120** to provide network access to various satellite terminals **130a-130c**. Each satellite terminal **130a-130c** can be a VSAT that provides network access to one or more client devices **140a-140c** (e.g., phones, laptop computers, desktop computers, Internet of Things (IoT) devices, appliances, etc.). The gateway **110** connects to a network **150**, such as a wide area network, core network, or other public network such as the Internet. Various servers **160** and other devices can communicate with the client devices **140a-140c** through the network connections that the system provides.

[0048] In the example of FIG. 1, the gateway **110** includes a transmitter subsystem **112** and a receiver subsystem **114**, as well as a status monitoring system **116**. The transmitter subsystem **112** and the receiver subsystem **114** may be configured to use frequencies in different bands of the electromagnetic spectrum. For example, the transmitter subsystem **112** used for uplink to the satellite **120** can transmit data in multiple channels in the V band. The receiver subsystem **114** may be configured to receive data transmitted in channels in the Q band. This may be reversed if desired (e.g., transmit in the Q band, receive in the V band). Of course, the gateway **110** may optionally be configured to transmit and receive in the same band or bands also (e.g., transmit and receive in the Q band, transmit and receive in the V band, transmit and receive in the Ka band, etc.). More generally, the gateway **110** can be designed to operate (e.g., transmit and/or receive) in the extremely high frequency (EHF) range designation of the International Telecommunication Union (ITU), and in the Ka band (26.5-40 GHz) and/or V band (40-75 GHz) defined by the Institute of Electrical and Electronics Engineers (IEEE), and/or the Q band (e.g., 36-46 GHz) defined by the International Organization for Standardization (ISO).

[0049] The intermediate frequencies used by the gateway **110** are often in the L band (e.g., 1-2 GHz) and S band (e.g., 2-4 GHz), and frequencies in the C band (e.g., 4-8 GHz) may also be used.

[0050] Various configurations of the transmitter subsystem **112** and receiver subsystem **114** are described with respect to the subsequent figures. The gateway **110** can be configured to

communicate with a satellite **120** in low earth orbit (LEO) (e.g., having an orbital period of about 128 minutes or less), in medium earth orbit (MEO) (e.g., a period of about 128 minutes to 24 hours), or in a higher orbit.

[0051] As discussed further below, the gateway **110** and the satellite **120** can be configured to communicate using multiple radiofrequency channels. For example, multiple channels that each have a predetermined bandwidth (e.g., 500 MHz) can be defined for uplink and for downlink. The frequency channels defined for uplink communication can be different from and non-overlapping with the frequency channels defined for downlink communication. For example, uplink and downlink use separate frequency ranges, and can be separated by a predetermined amount of bandwidth, such as 0.5 GHz, 1 GHz, 2 GHz, 3 GHz, etc. In some implementations, the uplink and downlink channels are separated by at least 4 GHz. For example, the lower end of the frequency range for the set of downlink channels can begin at least 4 GHz above the upper end of the frequency range for the set of uplink channels. This is only optional, however, and any appropriate non-overlapping frequency ranges can be set for uplink and downlink, with any appropriate level of separation between the ranges.

[0052] The gateway **110** can be configured to receive data in a total bandwidth (e.g., across all downlink channels) of at least 3 GHz, and in many cases a larger bandwidth (e.g., at least 4 GHz, 4.5 GHz, 5 GHz, 5.5 GHz, etc.). Using 5 GHz of reception bandwidth, the gateway **110** can provide ten 500 MHz channels for reception for each polarization (e.g., 10 RHCP downlink channels and 10 LHCP downlink channels). Other channel bandwidths may be used to split the overall reception bandwidth to provide more or fewer channels. For example, for the same 5 GHz of reception bandwidth, the gateway **110** can provide twenty 250 MHz channels per polarization or five 1.0 GHz channels per polarization.

[0053] The gateway **110** can be configured to transmit data in a total bandwidth (e.g., across all uplink channels) of at least 3 GHz, and in many cases a larger bandwidth (e.g., at least 4 GHz, 4.5 GHz, 5 GHz, 5.5 GHz, etc.). For example, using 4 GHz of transmission bandwidth, the gateway **110** can provide eight 500 MHz uplink channels per polarization (e.g., 8 RHCP uplink channels and 8 LHCP uplink channels). Of course, uplink channels can be defined with other bandwidths (e.g., 250 MHz, 1 GHz, etc.) to split uplink bandwidth into more or fewer channels.

[0054] FIG. 2 shows examples of frequency plans for uplink transmission and downlink transmission that can be used in the gateway **110**, such as an uplink frequency plan **300** and a downlink frequency plan **540**.

[0055] The transmitter subsystem **112** can use the uplink frequency plan **300**, which provides eight 500 MHz uplink channels. Signals for pairs of uplink channels are transmitted together, so that there are four different intermediate frequency signals IF1-IF4 provided to a quad-band upconverter **310**. Each of the four inputs to the upconverter **310** has a bandwidth of 1 GHz, and in the example, each IF signal IF1-IF4 spans a frequency range from 1.45 GHz to 2.45 GHz. The upconverter **310** shifts the IF signals IF1-IF4 to different portions of the V band. In the example, the upconverted output is provided at a range of 47.2 GHz to 51.4 GHz. With this configuration, there are four interfacility links (IFLs) that are active for each polarization used for transmission (e.g., one IFL for each of the intermediate frequencies IF1-IF4 for RCHP transmission, and one IFL for each of the intermediate frequencies IF1-IF4 for LHCP transmission). Each IFL carries 1.0 GHz of bandwidth. In some implementations, there are also for standby IFLs that can be switched to in the event of a failure for one of the primary IFLs. This configuration can provide a very small footprint for the transmitter subsystem components, such as providing all of the modulators in a single unit of rack space footprint.

[0056] The gateway **110** can use the uplink frequency plan **300** (or other uplink frequency plans discussed below) for multiple polarizations concurrently. For example, the frequency plan **300** can be used to transmit right hand circular polarized (RHCP) signals, and the same frequency plan **300** can also be used to concurrently transmit left-hand polarized (LHCP) signals using the same

channel definitions and intermediate frequency ranges.

[0057] In the example of FIG. 2, the receiver subsystem **114** uses a downlink frequency plan **540**. This involves receiving radiofrequency (RF) input in the Q band. For example, the receiver subsystem **114** can receive signals in a range from 37.5 GHz to 42.5 GHz. The RF input is provided to a tri-band downconverter **550**, which down converts to three intermediate frequency outputs IF1-IF3. In this example, IF1 has a bandwidth of 2.0 GHz, IF2 has a bandwidth of 2.0 GHz, and IF3 as a bandwidth of 1.0 GHz. IF1 and IF2 each include signals in the range from 2.0 GHz to 4.0 GHz, and IF3 as signals in the range from 2.0 GHz to 3.0 GHz.

[0058] The gateway **110** can use the downlink frequency plan **540** (or other downlink frequency plans discussed below) for multiple polarizations concurrently. For example, the frequency plan **540** can be used to receive right hand circular polarized (RHCP) signals, and the same frequency plan **540** can also be used to concurrently receive left-hand circular polarized (LHCP) signals using the same channel definitions and intermediate frequency ranges. In this configuration, there are nine IFLs, for example, three RHCP IFLs, three LHCP IFLs, and three standby IFLs that can be allocated or switched into use in place of the others in the event of a failure.

[0059] The frequency plans shown in FIG. 2 demonstrate how the gateway **110** can be configured to use an uplink frequency plan and a downlink frequency plan that have various different characteristics, such as differences in the number of bands, the intermediate frequencies used, and the band of the electromagnetic spectrum from which the RF signals are placed. Other variations are described below. For example, the gateway **110** can be configured to use any of the different uplink frequency plans **300**, **350** (FIGS. 3A-3B) with any of the downlink frequency plans **500**, **520**, **540**, **560** (FIGS. 5A-5D). Similarly, the gateway **110** can be configured to implement the transmitter subsystem **112** using any of the transmitter subsystems **400a-400f** (FIGS. 4A-4F), and can be configured to implement the receiver subsystem **114** using any of the receiver subsystems **600a-600g** (FIGS. 6A-6G).

[0060] FIGS. 3A and 3B show different uplink frequency plans **300**, **350** that can be used to implement upconverters in the transmitter subsystem **112** of the gateway **110**. Either of the uplink frequency plans **300**, **350** can be used to implement the transmitter subsystem **112** for uplink communication with the satellite **120**.

[0061] FIG. 3A shows an uplink frequency plan **300** in which signals to be transmitted are provided in four separate inputs. This results in four IFLs per polarization. The signals transmitted on each IFL have a 1.0 GHz bandwidth, each having signals to be transmitted in an intermediate frequency range from 1.45 GHz-2.45 GHz. As discussed above, the frequency plan **300** uses a quad-band upconverter **310**, which provides up converted output in the V band. In this case, the output spans the range from 47.2 GHz to 51.4 GHz.

[0062] The quad-band upconverter **310** includes four mixers **302**, one mixer for each input IFL. The mixers **302** each use different local oscillator signals to distribute the signals from the four different IFLs to adjacent, non-overlapping intermediate frequency regions. The results of the mixing are provided to a summing element **304**, then adjusted with an adjustable gain element **306** as well as other signal processing components (e.g., amplifiers, filters, gain slope compensation, temperature compensation, and other elements). The combined signal, as adjusted with the adjustable gain elements **306** (which labeled with "PC" to represent power control inputs) is then provided to the frequency mixer **308** which upconverts the signals up to the desired frequency range in the V band. The quad-band upconverter **310** can also include filters, amplifiers, compensation elements, and so on after the frequency mixer **308**.

[0063] FIG. 3B shows an alternative uplink frequency plan **350**, in which there are three IFLs per polarization providing input signals to be transmitted. The first two IFLs each carry IF signals that have a bandwidth of 1.5 GHz (e.g., signals for three 500 MHz channels), in the frequency range from 0.95 GHz to 2.45 GHz. The third IFL receives input signals with the bandwidth of 1.0 GHz, in a frequency range from 1.45 GHz to 2.45 GHz. As another option, the signals for IFL3 (e.g.,

signals for two 500 MHz channels) may be provided in a frequency range from 0.95 GHz to 1.95 GHz.

[0064] The frequency plan **350** uses a tri-band upconverter **360** that uses three mixers **302**, and operates in a similar manner to the upconverter **310**. For example, a summing element **304** combines the mixer output from the mixers **302**, a gain adjustment element **306** or other components adjusts the signal further, and a final mixer **308** provides the signals in the desired range, for example, from 47.2 GHz to 51.4 GHz.

[0065] FIGS. **4A-4F** show different example transmitter subsystems **400a-400f** that can be used to implement the transmitter subsystem **112** of the satellite gateway **110**. Each example includes a network switch **402**, modulators **410**, an intermediate frequency distribution network **420**, upconverters **430**, amplifiers **440**, radio frequency switches **450**, and a spectrum monitoring module **460**. The function of the network switch **402**, amplifiers **440**, radiofrequency switches **450**, and spectrum monitoring module **460** are generally consistent across the various examples. The modulators **410**, IF distribution network **420**, and upconverters **430** can have different configurations or different sets of features that vary from one example to the next. For example, different types of modulators **410** are labeled as modulators **410a-410f**, different types of IF distribution networks **420** are labeled as IF distribution networks **420a-420f**, and different types of upconverters **430** are labeled as upconverters **430a-430f**.

[0066] In the example, each switch **450** is a 4-port device, with two input ports and two output ports, sometimes referred to as a “baseball switch.” This type of switch is configured with four radiofrequency paths. For example, each switch **450** can be a transfer relay that has two independent RF input paths that are transferred simultaneously between the two output ports when switching occurs. This device has two states: (1) a first state in which input port **1** is coupled to output port **1** and input port **2** is coupled to output port **2**, and (2) a second state in which input port **1** is connected to output port **2** and input port **2** is coupled to output port **1**. The switching action is similar to a dual-pole double throw (DPDT) switch, except the two output ports are commonly shared with each input port (e.g., rather than each input port having a separate pair of output ports, the devices switches between only two output ports). Each switch **450** can configured as a relay or other electrically-actuated device to enable the controller to control the state of each switch **450** electrically.

[0067] FIG. **4A** shows an example configuration for a transmitter subsystem **112** that can be used to provide various different amounts of bands or channels, as well as different levels of redundancy. Either of the frequency plans **300**, **350** of FIGS. **3A** and **3B** can be used with the example of FIG. **4A**. In the example, there are several different variables that can be set in the system, including variables m , n , and i . The variable i represents an integer number of bands that each block converter is configured to process. For example, the value of i may be 4 when using the frequency plan **300**, or may be three when using the frequency plan **350**. The values of m and n represent the number of modulator outputs in each of two groups **411**, **412** of modulators **410a**.

[0068] In FIG. **4A**, a transmitter subsystem **400a** receives data from a LAN to transmit using the connection with the satellite **120**. The received data is processed by a network switch **402** and potentially other packet processing hardware. The network switch **402** divides the data to be transmitted among the various uplink channels and routes the digital data to the appropriate modulators **410a**. The modulators **410a** are shown in two groups. A primary group **411** is designated to handle modulation for RHCP transmission and for LHCP transmission. There are m modulator output lines providing intermediate frequency outputs to the IF distribution network **420a**. For example, with each polarization having i different bands or IFs, the value of m can be $2i$. In other words, when using tri-band upconverters (e.g., $i=3$), the value of m can be six (e.g., $m=3 \times 2=6$).

[0069] A secondary group **412** of modulators **410a** is placed in standby mode, providing redundant modulator capacity that is ready to be used in the event of a failure on one of the modulators **410a**

in the primary group **411**. For the primary modulators **410a**, there are m different modulator output lines or ports, and for the secondary modulators **410a** there are n output lines or ports. The ratio between m and n indicates the level of redundancy among the modulators **410a**. In some cases, m and n may be the same, and a ratio of 1:1 provides the same amount of standby or reserve capacity equal to the number of primary modulators. The outputs of the modulators **410a** are intermediate frequency signals having the data for transmission modulated onto the various channels defined for uplink, with different data to be transmitted modulated for the respective bands used in the frequency plan. For example, if the uplink frequency plan **300** is being used for transmission, there will be four primary outputs for each polarization, with each output including modulated signals in the intermediate frequency range of 1.45 GHz to 2.45 GHz (e.g., for two 500 MHz channels). [0070] Modulated output signals from the modulators **410** are provided to the IF distribution network **420A**, which in this example is an intermediate frequency IF switch matrix **421**. The switch matrix **421** has two sets of outputs, each carrying i number of separate connectors. For example, with the frequency plan **300**, there will be i=4 signals transmitted for the RHCP section and also i=4 signals transmitted for the LHCP section. The switch matrix **421** has the capability to route each of the outputs of the primary modulators in the group **411** to either of the upconverters **430a**. Similarly, the switch matrix **421** has the capability to route each of the outputs of the secondary modulators in the group **412** to either of the upconverters **430a**. As a result, the switch matrix **421** can selectively connect any output of a modulator **410a** with at least one input of the first upconverter **430a-1** or at least one input of the second upconverter **430a-2**. The switch matrix **421** can additionally be capable of connecting each modulator output line to a termination that is not provided to any input of an upconverter **430a**.

[0071] Thus, any of the inputs to the switch matrix **421** can be routed to either upconverter **430a**, even across the different polarizations. In other words, any of the modulator outputs, whether in the group for RHCP or LHCP, can be switched to be part of any of the two sets of i outputs to different upconverters **430a**. This switching can be provided for modulators **410a** in the standby or secondary group **412** as well as for the modulators **410a** in the primary group **411**. For example, if one-to-one redundancy is desired for both polarizations, there may be eight primary modulator outputs (e.g., m=8) and a corresponding eight secondary modulator outputs (e.g., n=8), for a total of 16 inputs to the switch matrix **421**. In this scenario, i equals 4, and there are a total of eight outputs of the switch matrix **421**. For each of the eight modulator outputs, the switch matrix **421** can be set to provide the modulator output to an input port of either of the two upconverters **430a**.

[0072] In some implementations, the switch matrix **421** is arranged to be able to route any of the modulator outputs to any of the upconverter inputs. For example, when i=4, m=8, and n=8, [0073] the switch matrix **421** can be arranged so that any of the 16 modulator outputs provided as input to the switch matrix **421** can be routed to any of the eight outputs of the switch matrix **421**. This provides a great deal of versatility and handling modulator signals and dealing with any potential failures in the modulators **410a**.

[0074] Output of the distribution network **420a** is provided to the upconverters **430a** (labeled "BUC" for block upconverter). Each of these upconverters **430A** can be one of the upconverters **310** or **360**, according to the frequency plan **300**, **350** selected to be used. As discussed above, the upconverters **430a** accept multiple intermediate frequency signals on different IFLs, with each IFL carrying data for one or more channels. The upconverters **430a** each up-convert the signals on different IFS to different portions of the V band.

[0075] The output of each upconverter **430a**, containing the upconverted signals for each of the multiple inputs to the upconverter **430a**, are then amplified using one of the amplifiers **440**.

[0076] Output from the amplifiers **440** is then provided to a set of radiofrequency switches **450**. The switches **450** are arranged so that either of the amplifier outputs can be provided to either of the antenna feeds **451** or **452** or can be connected to a termination instead. This provides additional flexibility to transmit the output of either amplifier on either antenna feed **451**, **452** or neither of

them.

[0077] The transmitter subsystem **400a** can include a spectrum monitoring module **460** that receives signals at multiple points from taps or ports. These monitored points can include each of the connections to the upconverters **430a** so that the intermediate frequency signals output from the IF switch matrix **421** are monitored. In addition, the module **460** can receive outputs from each of the amplifiers **440**, as well as signals provided on each of the antenna feeds **451** and **452**.

[0078] The spectrum monitoring module also includes a connection **461** for sending control signals and receiving status information from the switch **402** or other network traffic processing hardware. For example, based on the characteristics of signals on one of the input lines to the upconverters **430A**, the module **460** may determine that signals on one of the lines are abnormal or absent. From this, the module **460** can determine that a failure has occurred in one of the modulators **410** or that the switch matrix **421** is not providing the correct output. The module **460** can store the data indicating the state of the switch matrix **421** and/or the settings of the switch matrix **421** can be retrieved through a data and control connection **462**. With this information, the spectrum monitoring module **460** can identify which modulator and/or switching path is resulting in the abnormal or absent signals that were detected. The spectrum monitoring module **460** can provide information about the status of monitored lines to an external control system over the LAN.

[0079] The module **460**, or another external monitoring system, can also instruct the network switch **402**, modulators **410a**, and distribution network **420a** to route data for transmission to, and modulate using that data with, a different modulator **410a**. The instructions also can change the switch matrix settings so that the newly selected modulator **410a** has its output transmitted to the appropriate switch matrix output line on which the problem was detected. The other signals monitored by the module **460** can be used to verify proper operation of the rest of the transmitter subsystem **400a**. The module **460**, or an external control system, can also include a control connection with the switches **450** to be able to change the settings to route the different power amplifier **440** outputs to the appropriate antenna feed **451**, **452** or to a termination (labeled "T").

[0080] FIG. 4B shows another example transmitter subsystem **400b** that can be used in the transmitter subsystem **112**. This example uses quad-band upconverters **430b** and an IF distribution network **420b** that provides a direct connection between outputs of the modulators **410b** and the upconverters **430b** (e.g., without switching elements that would alter the coupling of modulator outputs to upconverter inputs).

[0081] In further detail, there is one modulator **410b** for each of two polarizations, e.g., one modulator for data to be transmitted using RHCP (e.g., MOD-R) and one modulator for data to be transmitted using LHCP (e.g., MOD-L). This results in four output lines per modulator **410b**, with each output line carrying a different intermediate frequency signal. The output lines of each modulator **410b** are provided to inputs of a corresponding upconverter **430b**.

[0082] Each modulator **410b** can be implemented as indicated as element **471**, with four outputs, each providing two channels of modulated data at a desired intermediate frequency (e.g., IF signals in the range from 1.45 GHz to 2.45 GHz, thus carrying signals for two 500 MHz uplink channels). The example shows how a modulator **410b** can include multiple digital-to-analog converters (DACs) which are used to produce multiple intermediate frequency output signals (e.g., four for each modulator **410b**). The amplifiers **440**, switches **450**, and spectrum monitoring module **460** operate as described above for FIG. 4A, except that without any switch matrix the monitoring module **460** does not need to read settings from or alter settings from components in the IF distribution network **420b**.

[0083] FIG. 4C shows another example transmitter subsystem **400c**, which is similar to the transmitter subsystem **400b** of FIG. 4B, but the IF distribution network **420c** adds switching capability not present in the distribution network **420b**. In particular, between the modulators **410c** and the upconverters **430c**, there are four switches **422**, configured to work at the intermediate frequency range of 1.45 GHz to 2.45 GHz, that can switch outputs of the different modulators **410c**

between the different upconverters **430c**. For example, a switch **422** can be coupled to receive input for the first IFL of the RHCP modulator **410c-1** and input of the first IFL of the LHCP modulator **410c-2**. The switch **422** can set whether (1) the RHCP first IFL is provided to the RHCP upconverter **430c-1** and the LHCP first IFL is provided to the LHCP upconverter **430c-2**, or (2) cross-switching occurs so that the RHCP first IFL is provided to the LHCP upconverter **430a-2** and the LHCP first IFL is provided to the RHCP upconverter **430a-1**.

[0084] The switches **422** provide the option for cross-switching, which can achieve improved availability during a double failure. For example, if one of the modulators **410c** experiences a failure and one of the upconverters **430c** or amplifiers **440** also experiences a failure, the spectrum monitoring module **460** can route signals from the functioning modulator **410c** to the functioning upconverter **430c** and amplifier **440**. The output from that amplifier **440** can then be routed to either of the antenna feeds **451**, **452** using an appropriate setting for the switches **450**.

[0085] FIG. **4D** shows another example of a transmitter subsystem **400d**. In this example, modulators **410d** providing three intermediate frequency outputs are used, and tri-band upconverters **430d** are used, according to the frequency plan **350**. The transmitter subsystem **400d** is similar to that of **400b**, in that the IF distribution network **420d** provides direct connections between modulators **410c** and upconverters **430d** without switching elements between them.

[0086] The modulators **410d** can each be implemented as illustrated as shown for modulator **472** to include multiple intermediate frequency generating components. For example, a modulator (“MOD”) can be coupled with two digital-to-analog converters (“DACs”), and each DAC can be coupled to two intermediate frequency generating components (“IFCs”) The outputs of pairs of IFCs that generate output in different frequency ranges can be summed together with a summing element (“Σ”). The summed output is provided to one of two output lines, as selected by a switch. For example, the top DAC is used to generate modulated output for two ranges of intermediate frequency output that can be summed together. Then, using a switch, the summed output can be provided to either a primary output (“1P”) or, if desired, provided instead to a secondary or standby output (“1S”). In the example of FIG. **4D**, however, the switches of the modulators **410d** are not used, and the secondary outputs, e.g., 1S and 2S, are not connected or used. Instead, for RHCP channels, there are two modulators **410d** and only the 1P and 2P outputs of the first RHCP modulator **410d** are used and only the 1P output of the second RHCP modulator **410d** is used, for a total of three IFLs or output lines for RHCP signals routed to the RHCP upconverter **430d**. Similarly, there are only three IFLS or output lines from the LHCP signals routed to the LHCP upconverter **430d**.

[0087] FIG. **4E** shows an example transmitter subsystem **400e** that has features as described for the transmitter subsystem **400d** of FIG. **4D**, but adds switches **422** in the IF distribution network **420e**. These switches **422** allow routing of modulator outputs from either the group **414** of RHCP modulators or the group **415** of LHCP modulators to either of the upconverters **430e**

[0088] As discussed for receiver subsystem **400d**, the modulators **410e** include switches that can switch between primary outputs (labeled 1P and 2P) and secondary outputs (labeled 1S and 2S). However, the switches of the modulators **410d** are not used, and the secondary outputs, e.g., 1S and 2S, are not connected or used.

[0089] FIG. **4F** shows another example transmitter subsystem **400f**. This example includes tri-band upconverters **430f**, of the type shown as upconverter **360** in FIG. **3B**. The receiver subsystem **400f** uses the uplink frequency plan **350** of FIG. **3B**.

[0090] In the transmitter subsystem **400f**, the modulators **410f** have features as shown in modulator implementation **472**. This includes switches that can switch between primary outputs (e.g., 1P and 2P) and secondary outputs (e.g., 1S and 2S). Three primary modulator outputs can be provided for each polarization (e.g., 1P and 2P for a first modulator and 1P of a second modulator). Three secondary modulator outputs can be provided for each polarization (e.g., 1S and 2S for a first modulator and 1S of a second modulator). This results in three primary IF outputs for RHCP, three

standby output lines for RHCP, three primary outputs for LHCP, and three secondary outputs for LHCP, for a total of 12 IFLs or output lines.

[0091] The IF distribution network **420f** provides passive distribution of modulator outputs from RHCP modulator group **414** to both upconverters **430f** and passive distribution of modulator outputs from LHCP modulator group **415** to both upconverters **430f**. Modulator output lines are coupled to summing elements **425** in the IF distribution network **420f**. This provides a combination of the primary RHCP outputs with the secondary LHCP outputs. In addition, the summing elements **425** combine the primary LHCP outputs with the secondary RHCP outputs. Using the switches shown in the modulator implementation **472**, the system can select to provide modulated output signals over (1) primary IFLs (e.g., the “P” lines, including 1P, 2P, etc.), or (2) secondary IFLs (e.g., the “L” lines, including 1S, 2S, etc.). The summing elements **425** combine the modulated signals of pairs of IFLs across modulator groups **414**, **415**. For example, a first RHCP primary output 1P is summed with a first LHCP secondary output 1S; a first LHCP primary output 1P is summed with a first RHCP secondary output 1S; and so on. When switches are set to use primary outputs, the secondary outputs are not connected, so the provided on the primary IFLs reach the corresponding upconverters **430f**. When the switches are set to provide output on the secondary output lines, there is no output on the primary lines, and cross-switching occurs (e.g., output from LHCP modulators **415** is provided to the RHCP upconverter **430f-1**, and output from the RHCP modulators **414** is provided to the LHCP upconverter **430f-2**. In this manner, the distribution network **420f** can route signals from either or both groups **414**, **415** of modulators **410f** to either of the upconverters **430f**, without any switching in the distribution network **420f**.

[0092] The arrangement of transmitter subsystem **400f** provides improved availability during a double failure, such as a failure in one or more modulators **410f** and a failure in one or more upconverters **430f** and/or amplifiers **440**. The arrangement involves no external switches, only passive combining of the different modulator outputs, and each modulator **410f** selects which output ports to use, from among the primary (P) and secondary(S) output ports.

[0093] FIGS. 5A-5D show different options of downlink frequency plans **500**, **520**, **540**, **560** for the receiver subsystem **114**. FIGS. 5A-5C each involve three bands (e.g., three different intermediate frequency outputs carrying different data) and use a tri-band downconverter. FIG. 5D involves four bands and uses a quad-band downconverter. FIGS. 5A and 5B each receive Q-band signals in a range from 37.5 GHz to 42.0 GHz, which can be divided into nine 500 MHz channel segments. FIGS. 5C and 5D receive Q-band signals in a range from 37.5 GHz to 42.5 GHz, which can be divided into ten 500 MHz channel segments.

[0094] FIG. 5A shows a downlink frequency plan **500**, which makes use of a tri-band downconverter **510**. The downconverter **510** includes a frequency mixer **502**, a divider **504**, and three mixers **506**. The downconverter **510** is configured to receive Q-band signals in the range from 37.5 GHz to 42.0 GHz and to down convert these to three intermediate frequency outputs, labeled IF1-IF3. The outputs IF1 and IF2 are each in a range from 2.0 GHz to 4.0 GHz, thus having a bandwidth of 2 GHz each. The output IF3 is in a range from 2.0 GHz to 2.5 GHz, for a bandwidth of 0.5 GHz.

[0095] The downlink frequency plan **500** can be used for multiple polarizations. For example, one tri-band downconverter **510** can be used for right hand polarized signals and another tri-band downconverter **510** can be used for left hand polarized signals. In addition, at least one additional downconverter can be provided as a redundant standby component that is able to take over for one of the other downconverters **510** in the event of a failure. With three downconverters **510** (e.g., RHCP, LHCP, and standby), this plan **500** would include 9 IFLs in total.

[0096] The example of FIG. 5A is optimized to limit the requirement of demodulator processing power and also the fewest number of demodulator is needed. It also can be implemented in the smallest footprint (e.g., physical space) of the four downlink frequency plan options in FIGS. 5A-5D.

[0097] FIG. 5B shows another example downlink frequency plan **520**. This example also uses a tri-band downconverter **530**. The downconverter **530** receives Q-band signals in the range from 37.5 GHz to 42.0 GHz and provides three intermediate frequency outputs IF1-IF3. In this example, each of the intermediate frequency outputs IF1-IF3 uses the same frequency range (e.g., 0.95 GHz to 2.45 GHz) and bandwidth (e.g., 1.5 GHz).

[0098] The downlink frequency plan **520** can be used for multiple polarizations. For example, one tri-band downconverter **530** can be used for right hand polarized signals and another tri-band downconverter **530** can be used for left hand polarized signals. In addition, at least one additional downconverter can be provided as a redundant standby component that is able to take over for one of the other downconverters **530** in the event of a failure. With three downconverters **530** (e.g., for RHCP, LHCP, and standby), this frequency plan **540** would include 9 IFLs in total.

[0099] FIG. 5C shows another example downlink frequency plan **540**. The example makes use of a tri-band downconverter **550** that receives Q band input signals in a range from 37.5 GHz to 42.5 GHz and produces output of three intermediate frequency outputs, IF1, IF2, IF3. In this example, the intermediate frequency outputs IF1 and IF2 are each in a range from 2.0 GHz to 4.0 GHz, thus having a bandwidth of 2.0 GHz each. The third intermediate frequency output IF3 is in a range from 2.0 GHz to 3.0 GHz for a bandwidth of 1.0 GHz.

[0100] The downlink frequency plan **540** can be used for multiple polarizations. For example, one tri-band downconverter **550** can be used for right hand polarized signals and another tri-band downconverter **550** can be used for LHCP signals. In addition, at least one additional downconverter can be provided as a redundant standby component that is able to take over for one of the other downconverters **550** in the event of a failure. With three downconverters **550** (e.g., RHCP, LHCP, and standby), this plan **540** would include 9 IFLs in total.

[0101] FIG. 5D shows another example downlink frequency plan **560**. This example uses a quad-band downconverter **570**. After the mixer **502**, the mixer output is divided by divider **504** that splits the signal four ways. The downconverter **570** includes four mixers for a 6 to create four intermediate frequency outputs, labeled IF1-IF4. The downconverter **570** is configured to receive Q band input in the range from 37.5 GHz to 42.5 GHz. The first three intermediate frequency outputs IF1, IF2, IF3 are each in a frequency range of 0.95 GHz to 2.45 GHz, each having a bandwidth of 1.5 GHz. The fourth intermediate frequency output IF4 has a frequency range of 0.95 GHz to 1.45 GHz, for a bandwidth of 0.5 GHz.

[0102] The downlink frequency plan **560** can be used for multiple polarizations as well as to provide a redundant standby capability. For example, one downconverter **570** can be used for RHCP signals, a second downconverter **570** can be used for LHCP signals, and a third downconverter **570** can be used as a redundant standby element. This results in four output lines per downconverter, for a total of 12 IFLs.

[0103] FIG. 6A to 6G show examples of different receiver subsystems **600a-600g** that can be used in the receiver subsystem **114** of the gateway **110** of FIG. 1. The example of FIG. 6A can use any of various different types of downconverters, including tri-band downconverters, quad-band downconverters, 5-band downconverters, and so on. The examples of FIGS. 6B, 6D, 6F, and 6G use tri-band downconverters. The examples of FIGS. 6C and 6D use quad-band downconverters, such as the downconverter **570** from FIG. 5D.

[0104] Each of the receiver subsystems **600a-600g** have certain general characteristics in common. For example, each receives signals in the Q band on an antenna feed **601** for RHCP signals and an antenna feed **602** for LHCP signals. The antenna feeds **601**, **602** are provided to radiofrequency switches **610**. These switches **610** can be 4-port devices, e.g., a baseball switch or transfer relay, as discussed above for the switches **450**. Each switch **610** is connected so that it is able to selectively (1) route the received signal to a primary amplifier **620** and downconverter **630** for the corresponding polarization or (2) route the input to a standby low noise amplifier **620** and standby downconverter **630**. In the receiver subsystems **600a-600g**, there is one standby amplifier **620** and

one standby downconverter **630** for the pair of polarizations. The standby components can be used in the event of a failure in either of the sets of primary components, but for only one polarization at a time. The output of each downconverter **630** includes multiple intermediate frequency signal outputs which are provided over an IF distribution network **640** to a set of demodulators **650**. Output of the demodulators **650** is provided to a switch or other network packet processing hardware which then provides the received data over the LAN.

[0105] In the various receiver subsystems **600a-600g**, the radiofrequency input in each case is in a range of 37.5 GHz to 42.5 GHz. Nevertheless, the different receiver subsystems **600a-600g** vary in the number of bands used for the downconverters **630**, in the types of distribution networks **640** that are used, and in the number and types of demodulators **650** that are used.

[0106] FIG. **6A** shows an example receiver subsystem **600a**. Radiofrequency signals in the Q band as detected by the antenna are provided at antenna feeds **601** and **602**. The antenna feed **601** carries RHCP signals, and the antenna feed **602** carries LHCP signals. The antenna feeds **601** and **602** are coupled to switches **610** which allow signals from each antenna feed **601**, **602** to be routed to either of two low noise amplifiers **620**. There are three low noise amplifiers **620** and three corresponding downconverters **630a**. The top low noise amplifier **620** and downconverter **630a** are the primary components for receiving RHCP signals, and the lower low noise amplifier **620** and downconverter **630a** are the primary components for receiving LHCP signals. The middle low noise amplifier **620** and downconverter **630a** are standby elements that normally remain in an operating standby mode without processing incoming signals when there are no failures on the primary components. However, in the event of a failure in either of the primary components, either the RHCP signals or the LHCP signals can be routed to the standby components. this provides redundancy and the ability to quickly recover from a failure in either of the primary sets of components.

[0107] The downconverter **630a** are each configured to process signals from i number of bands, where i is an integer. For example, the value of i may be 3, 4, or 5. As a result, each downconverter **630a** provides i number of intermediate frequency outputs.

[0108] The intermediate frequency outputs from each of the downconverter **630a** are provided to an intermediate frequency distribution network **640a**. In this example, the IF distribution network **640a** includes an IF switch matrix **641** that is configured to be able to couple any of the input signals from the downconverter **620a** to any of various outputs of the switch matrix **641**. This capability provides high versatility and redundancy in the event of failures of amplifiers, downconverters **630a** and, demodulators **650a**.

[0109] The receiver subsystem **600a** includes a set of demodulators **650a**, which are shown in two groups, a first group **651** with m number of primary demodulators **650a** and a second group **652** of n number of secondary demodulators **650a**. Each demodulator **650a** processes 0.5 GHz to 2.0 GHz of bandwidth, depending on the implementation. For example, if there are two polarizations used and a tri-band downconverter **630a** for each polarization, the value of variable m would be 3. There are double the number of m connections from the switch matrix **641** to the primary modulators, e.g., three outputs for RHCP and three outputs for LHCP.

[0110] For the second group of demodulators **650a**, there are n number of outputs of the switch matrix **641**, which is set according to the level of redundancy desired. For example, if one to one redundancy is desired, n equals m. As another example, if redundancy is desired for only half the capacity of the group **651** of primary demodulators **650a**, then n can be half of m. The redundancy in the demodulators **650a** between the first group **651** and the second group **652** can be the same level of redundancy as with the downconverters **630a** and amplifiers **620**, but is not required to be. For example, the system can be designed to provide an amount of modulator redundancy (e.g., a number of standby modulators equal to the number of primary modulators) that is greater than redundancy for the primary downconverters **630a** or primary amplifiers **620** (e.g., one standby downconverter **630a** and standby amplifier **620** for every two primary downconverters **630a** and primary amplifiers **620**). Optionally, for one-to-one redundancy for the modulators **650a**, the

number of secondary demodulators **650a** and the number of links to the secondary demodulators **650a** from the switch matrix **641** can be the same as the number of primary demodulators **650a** and corresponding links to the switch matrix **641**.

[0111] As noted above, the configuration of receiver subsystem **600a** can be set to make use of various different bands and types of downconverters. Examples of these configurations are shown in FIG. 7.

[0112] FIG. 7 shows four different examples of frequency and downconverter configurations that can be used in the receiver subsystem **600a** of FIG. 6A.

[0113] The first configuration **710** uses tri-band downconverters, where the intermediate frequency outputs have a bandwidth of 2.0 GHz, 2.0 GHz, and 0.5 GHz, respectively, or use a bandwidth of 2.0 GHz, 2.0 GHz, and 1.0 GHz, respectively. The downconverters are configured to provide intermediate frequency signals in a range of 2.0 GHz to 4.0 GHz. In the configuration **710**, with a tri-band downconverter, $i=3$. In this example, m equals 5, so 5 primary demodulators are used, and n secondary demodulators are used where the value of n can be a value of at least 1, and up to 5, to provide redundancy. This results in nine (e.g., $3 \times i$) input lines to the switch matrix **641**, and from 12 to 20 output lines from the switch matrix **641** depending on the level of demodulator redundancy desired.

[0114] The second configuration **720** uses tri-band downconverters, where the intermediate frequency outputs have a bandwidth of 1.5 GHz, 1.5 GHz, and 1.5 GHz, respectively. The downconverters are configured to provide intermediate frequency signals in a range of 2.0 GHz to 4.0 GHz. In the configuration **720**, with a tri-band downconverter, $i=3$. In this example, m equals 6, so 6 primary demodulators are used, and n secondary demodulators are used where the value of n can be a value of at least 1, and up to 6, to provide redundancy. This results in nine (e.g., $3 \times i$) input lines to the switch matrix **641**, and from 14 to 24 output lines from the switch matrix **641** depending on the level of demodulator redundancy desired.

[0115] The third configuration **730** uses quad-band downconverters, where the intermediate frequency outputs have a bandwidth of 1.5 GHz, 1.5 GHz, 1.5 GHz, and 0.5 GHz, respectively. The downconverters are configured to provide intermediate frequency signals in a range of 0.95 GHz to 2.45 GHz. In the configuration **730**, with a quad-band downconverter, $i=4$. In this example, m equals 7, so 7 primary demodulators are used, and n secondary demodulators are used where the value of n can be a value of at least 1, and up to 7, to provide redundancy. This results in twelve (e.g., $3 \times i$) input lines to the switch matrix **641**, and from 16 to 28 (e.g., $2m+2n$) output lines from the switch matrix **641** depending on the level of demodulator redundancy desired.

[0116] The fourth configuration **740** uses quad-band downconverters, where the intermediate frequency outputs have a bandwidth of 1.0 GHz, 1.0 GHz, 1.0 GHz, 1.0 GHz, and 0.5 GHz, respectively. The downconverters are configured to provide intermediate frequency signals in a range of 0.95 GHz to 2.45 GHz. In the configuration **740**, with a 5-band downconverter, $i=5$. In this example, m equals 5, so 5 primary demodulators are used, and n secondary demodulators are used where the value of n can be a value of at least 1, and up to 5, to provide redundancy. This results in fifteen (e.g., $3 \times i$) inputs to the switch matrix **641**, and from 12 to 20 (e.g., $2m+2n$) outputs from the switch matrix **641** depending on the level of demodulator redundancy desired.

[0117] Referring again to FIG. 6A, the status monitoring module **260** receives monitoring inputs at various points in the receiver subsystem **600a**. For example, the status monitoring module **660** can receive and monitor signals at the antenna feeds **601**, **602**, and also receive signals on the transmission lines from the downconverters **630a** to the switch matrix **641** in the distribution network **640a**. In addition, the status monitoring module **660** includes the capability to provide output to various points in the receiver subsystem **600a**. For example, the status monitoring module can provide an output on a monitoring line **605** to provide test signals to the standby low noise amplifier **620** and downconverter **630a**, and then can monitor the result of the input test signals with the connection **606**. In addition, the status monitoring module **660** can be capable of

transmitting signals to the antenna feeds **601**, **602** to perform additional testing and monitoring of the primary amplifiers **620** and primary downconverters **630a**.

[0118] The status monitoring module **660** also includes the capability to adjust the settings for the switches **610** and the switch matrix **641** of the IF distribution network **640a**. This capability enables the status monitoring module **662** change the settings of the switches **610**, so that either of the antenna feeds **601**, **602** (but not both at the same time) can be selectively coupled to the standby low noise amplifier **620** and standby downconverter **630a** (e.g., the middle path shown between the RHCP and LHCP paths). The status monitoring module **660** can also send control signals to the switch matrix **641** in the IF distribution network **640a** to route signals to the appropriate inputs of the demodulators **650a**. This allows the status monitoring module **662** respond to detected anomalies or failures and to change the routing of intermediate frequency output lines from the downconverter **630a** to the different input transmission lines to the demodulators **650a** by adjusting the settings of the switch matrix **641**. For example, if a downconverter **630a** or a low noise amplifier **620** fails, and the switches direct incoming RF signals to the standby amplifier **620** and downconverter **630a**, the status monitoring module **660** adjusts the switch matrix **641** settings to route the outputs from the standby downconverter to the primary demodulator that is used. As another example, if there is a failure in one of the primary modulators, the status monitoring module **660** detects the failure and then adjust the switch matrix **641** settings to route the inputs to the failed demodulator to one of the secondary demodulators instead.

[0119] In addition, FIG. **6A** also shows an example demodulator implementation **670** that may be used. In the demodulator implementation **670**, a demodulator **671** can have two different sections **672a-672b**, each with its own IF processing section **673** and ADC **674**. Each section **675a-675b** can have two IF inputs: (1) section **675a** has a primary input 1P and a secondary output 1S, and (2) section **675b** has a primary output 2P and a secondary output 2S. Each section **672a**, **672b** includes a switch **675a**, **675b** that selects between the two inputs to the section and couples only the selected input for demodulation. In this manner, the demodulator implementation **670** can provide switching ability to optionally share a single demodulator processing unit **671** among to intermediate frequency signals each selected from a different pair of input lines.

[0120] With the switch matrix **641**, the receiver subsystem **600a** provides very high versatility, as well as a level of redundancy that can vary according to the needs of the system.

[0121] Using the IF distribution network **640**, signals downconverted from both the RHCP and LHCP antenna feeds **601**, **602** can be routed to and analyzed by the demodulators **650a**, which allows the demodulators **650a** to provide cross polarization (XPOL) compensation.

[0122] FIG. **6B** shows another example receiver subsystem **600b**. The receiver subsystem **600b** includes the antenna feeds **601**, **602**, the RF switches **610**, and low noise amplifiers **620**. As discussed above, the receiver subsystem **600a** also includes three downconverters **630b**: a primary downconverter for the RHCP signals, a primary downconverter for the LHCP signals, and a standby downconverter. In this example, each of the downconverters **630b** is a tri-band downconverter. For example, the three bands can each have a bandwidth of 1.5 GHz. In some cases, the intermediate frequency output for each downconverter output is from 0.5 GHz to 2.45 GHz.

[0123] Each of the three downconverters **630b** has three output transmission lines or IFLs, for a total of 9 IFLs. The output lines of the downconverter **630b** are provided through an IF distribution network **640b** to a set of demodulators **650b**. Unlike the receiver subsystem **600a** of FIG. **6A**, the receiver subsystem **600b** does not used switching between the downconverter **630b** and the demodulators **650b**. Instead, direct connections are made between each downconverter **630b** and a corresponding set of demodulators **650b**. In other words, there are three different paths, one each for RHCP, LHCP, and standby, and after the routing of the switches **610**, signals are not changed from one path to another.

[0124] However, each IFL is passed to a divider **680** that splits receive signals into two sub-bands

or channels to be demodulated separately. For example, the demodulator implementation **670** can be used three times for each polarization, and at the front of each and demodulator implementation **670**, a divider can separate the 1.5 GHz intermediate frequency bandwidth into two channels, one with 1.0 GHz bandwidth and another with 0.5 GHz bandwidth.

[0125] In the receiver subsystem **600b**, the system uses a simple configuration of chain switching. Up to 2.0 GHz of bandwidth is used per IFL. Each demodulator processes 1.5 GHz of bandwidth. Three demodulators are used per path, for a total of nine demodulators.

[0126] FIG. **6C** shows another example receiver subsystem **600c**. The receiver subsystem **600c** has a structure and operation as described above for receiver subsystem **600b**, but uses quad-band downconverters instead of tri-band downconverters, and uses four demodulators per receive path instead of three.

[0127] In the receiver subsystem **600c**, up to 1.5 GHz is needed for IFL. Each demodulator processes 0.5 GHz of bandwidth to 1.5 GHz of bandwidth. Four demodulators **650c** are used per path, for a total of 12 demodulators.

[0128] Each uplink path (e.g., RHCP and LHCP) includes four demodulators **650c**, each implemented with the demodulator implementation **670**. A different IFL is coupled to demodulator in the group. Each IFL carries a different intermediate frequency signal in the range of 0.95 GHz to 2.5 GHz for IF1-IF3, and 0.95 GHz to 1.45 GHz for IF4. A divider is provided before each of the demodulators. For IF1-IF3, the divider divides the incoming signal into a 1 GHz component and a 0.5 GHz component to facilitate demodulation. For IF4, only a 0.5 GHz signal is used. Nevertheless, the capability of the system can be extended by providing additional signals on IF4 and dividing them to provide an extended band for demodulation. An advantage of the receiver subsystem **600c** is that passive elements are used in the distribution network **640c**, and so no active switches or switch control or processing is required.

[0129] FIG. **6D** shows an example of receiver subsystem **600d** used to receive downlink signals in the gateway **110**. In the example, a passive distribution scheme is used. The configuration shown in FIG. **6D** has the ability to provide XPOL compensation. In addition, the configuration allows for path selection by the demodulators **650d**. The system has six primary demodulators **650d** and can include, for example, three to six standby demodulators **650d**. The configuration makes use of a tri-band receiving process, in this case, a frequency arrangement in which each polarization has three intermediate signals each having a 1.5 GHz bandwidth. For example, each intermediate frequency signal can have a bandwidth from 0.95 GHz to 2.45 GHz. The IF distribution network **640d** includes a series of signal dividers **642**. Each of the signal dividers **642** receives an IF signal (e.g., one band of one downconverter **630d**) and splits the signal into two different frequency ranges, e.g., for different channels or groups of channels. Both output signals from the divider **642** are then provided to a demodulator in the group **655** as well as to a demodulator in the group **656**.

[0130] In the example, element **643** shows a collection of three signal dividers **642** used for the primary LHCP path. The same structure and connections can similarly be used for the dividers **642** for the standby path and also for the RHCP path. A different band of output (e.g., one IF signal) from the LHCP downconverter **630d** is provided to each divider **642**. Each divider **642** divides the signal into two different frequency ranges or channel groups, and also splits those signals so that a copy of each of divided signal is sent to the upper path and lower path. As a result, all six of the divided outputs are provided to the group **655** of primary demodulators **650d**, and all six of the divided outputs are also provided to the group **656** of secondary demodulators **650d**.

[0131] The demodulator implementation **670** shows how a demodulator **650d** can handle three incoming signals with two demodulator paths. For example, each demodulator **650d** has connections to three input signals: an RHCP signal on the 1P input, an LHCP signal on the 2P input, and a signal from the standby demodulator on both the 1S and 2S inputs. The switches **675a**, **675b** in the demodulator allow selection of either a primary input (RHCP or LHCP) or the standby input provided through the divider **681**.

[0132] FIG. 6E shows an example of a receiver subsystem **600e** that can be used to receive downlink signals in the gateway **110**. The receiver subsystem **600e** operates as discussed above for the receiver subsystem **600d**, but uses quad-band downconverters **630c** according to the frequency plan **560** of FIG. 5D. This results in a greater number of outputs from each downconverter **630c** (e.g., four instead of three for FIG. 6D), and also involves four dividers **642** instead of three. In the example, the total bandwidth downconverted by each downconverter is 5.0 GHz. As discussed in FIG. 5D, there are three intermediate frequency signals that each have a bandwidth of 1.5 GHz, and another that uses only 500 MHz of bandwidth. The 1.5 GHz-wide IF signals can each be divided into two different signals, e.g., one with a 1.0 GHz bandwidth and another with 0.5 GHz bandwidth. In many cases, the demodulation process may be more efficient or may provide easier routing when signals having only one or two of the 500 MHz channels are demodulated from each input. The fourth IF signal does not need to be divided, as it carries only one channel of information to be demodulated.

[0133] FIG. 6F shows an example of a receiver subsystem **600f** that can be used to receive downlink signals in the gateway **110**. The receiver subsystem **600f** operates as described above for the receiver subsystem **600b** of FIG. 6B, but uses different intermediate frequency ranges and bandwidths for IF1-IF3. For example the receiver subsystem **600f** uses the downlink frequency plan **500** of FIG. 5A, which has intermediate frequencies of 2.0-4.0 GHz for IF1, 2.0-4.0 GHz for IF2, and 2.0-2.5 GHz for IF3. By contrast, the receiver subsystem **600b** uses the downlink frequency plan **520** of FIG. 5B, in which all of IF1-IF3 are in the range of 0.95 GHz to 2.45 GHz.

[0134] FIG. 6G shows an example of a receiver subsystem **600g** that can be used to receive downlink signals in the gateway **110**. The receiver subsystem **600g** operates as described above for receiver subsystem **600d** of FIG. 6D, but uses a different set of bands.

[0135] In the receiver subsystem **600g**, each upconverter **630g** provides three IF outputs, which can be split into 2.0 GHz, 2.0 GHz, and 0.5 GHz, respectively, for receiving a total of 4.5 GHz (e.g., nine 500 MHz channels). The three IF outputs of each upconverter **630g** can alternatively be 2.0 GHz, 2 GHz, and 1.0 GHz, respectively, for receiving a total of 5.0 GHz (e.g., ten 500 MHz channels). The intermediate frequency signals are in the range from 2.0 GHz to 4.0 GHz. This arrangement is different from the intermediate frequencies used in the example of FIG. 4D, which have a frequency range of 0.95 GHz to 2.45 GHz and are split into three groups of 1.5 GHz, 1.5 GHz, and 1.5 GHz.

[0136] A number of embodiments have been described. Nevertheless, it will be understood that various modifications can be made without departing from the spirit and scope of the invention. In addition, the logic flows depicted in the figures do not require the particular order shown, or sequential order, to achieve desirable results. In addition, other steps can be provided, or steps can be eliminated, from the described flows, and other components can be added to, or removed from, the described systems.

[0137] A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. For example, various forms of the flows shown above may be used, with steps re-ordered, added, or removed.

[0138] Embodiments of the invention and all of the functional operations described in this specification can be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Embodiments of the invention can be implemented as one or more computer program products, e.g., one or more modules of computer program instructions encoded on a computer readable medium for execution by, or to control the operation of, data processing apparatus. The computer-readable medium can be a machine-readable storage device, a machine-readable storage substrate, a memory device, a composition of matter effecting a machine-readable propagated signal, or a combination of one or more of them. The term “data

processing apparatus” encompasses all apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, or multiple processors or computers. The apparatus can include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them. A propagated signal is an artificially generated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal that is generated to encode information for transmission to suitable receiver apparatus.

[0139] A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

[0140] The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit).

[0141] Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Moreover, a computer can be embedded in another device, e.g., a tablet computer, a mobile telephone, a personal digital assistant (PDA), a mobile audio player, a Global Positioning System (GPS) receiver, to name just a few. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

[0142] To provide for interaction with a user, embodiments of the invention can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube) or LCD (liquid crystal display) monitor, for displaying information to the user and a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input.

[0143] Embodiments of the invention can be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a

graphical user interface or a Web browser through which a user can interact with an implementation of the invention, or any combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network (“LAN”) and a wide area network (“WAN”), e.g., the Internet.

[0144] The computing system can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

[0145] While this specification contains many specifics, these should not be construed as limitations on the scope of the invention or of what may be claimed, but rather as descriptions of features specific to particular embodiments of the invention. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

[0146] Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

[0147] Particular embodiments of the invention have been described. Other embodiments are within the scope of the following claims. For example, the steps recited in the claims can be performed in a different order and still achieve desirable results.

Claims

1. A receiver system comprising: a set of downconverters comprising (i) a first downconverter to downconvert signals received for a first of multiple polarizations, and (ii) a second downconverter to downconvert signals received for a second of the multiple polarizations, wherein each of the downconverters is configured to provide multiple intermediate frequency outputs that overlap in frequency range and carry different data; a set of demodulators comprising at least one demodulator for each of the multiple polarizations, wherein each of the demodulators is configured to extract data from a different intermediate frequency signal; an intermediate frequency distribution network configured to distribute the multiple intermediate frequency output signals from each of the downconverters to the inputs of the demodulators; and a network interface configured to provide extracted data to a communication network.
2. The receiver system of claim 1, wherein each of the downconverters is configured to output three or four intermediate frequency signals based on an input radiofrequency signal.
3. The receiver system of claim 1, further comprising: a third downconverter; and a set of radiofrequency switches that are operable to (i) selectively switch radiofrequency input for the first of the multiple polarizations from the first downconverter to the third downconverter, and (ii) selectively switch radiofrequency input for the second of the multiple polarizations from the second

downconverter to the third downconverter.

4. The receiver system of claim 1, wherein the intermediate frequency distribution network comprises a switch matrix that is programmable to selectively provide intermediate frequency outputs from any of the downconverters to at least one demodulator in any of multiple different groups of demodulators.
5. The receiver system of claim 1, wherein the intermediate frequency distribution network comprises frequency dividers for each downconverter, the frequency dividers providing the output of a downconverter to each of multiple sets of demodulators.
6. The receiver system of claim 1, wherein the multiple polarizations include right-hand circular polarization and left-hand circular polarization.
7. The receiver system of claim 1, wherein the receiver system is configured to receive signals in the Q band, over a bandwidth of at least 4 GHz.
8. The receiver system of claim 1, wherein the receiver system is configured to receive data for a plurality of different channel segments, and each of the multiple intermediate frequency outputs of the downconverters includes modulated signals for multiple of the channel segments.
9. The receiver system of claim 8, wherein the receiver system is configured to receive signals for each of the multiple polarizations in multiple channel segments each having bandwidth between 250 MHz and 2 GHz.
10. A method for receiving and processing signals of multiple polarizations, the method comprising: receiving signals for a first and a second of multiple polarizations; downconverting the received signals for the first polarization using a first downconverter and the received signals for the second polarization using a second downconverter, wherein each downconverter providing multiple intermediate frequency outputs that overlap in frequency range and carry different data; distributing the multiple intermediate frequency outputs from each of the downconverters to a set of demodulators using an intermediate frequency distribution network; extracting data from different intermediate frequency signals using at least one demodulator for each of the multiple polarizations; and providing the extracted data to a communication network via a network interface.
11. The method of claim 10, wherein each of the first and second downconverters outputs three or four intermediate frequency signals based on an input radiofrequency signal.
12. The method of claim 10, further comprising: downconverting received signals using a third downconverter; and selectively switching radiofrequency input for the first polarization from the first downconverter to the third downconverter and for the second polarization from the second downconverter to the third downconverter using a set of radiofrequency switches.
13. The method of claim 10, wherein distributing the intermediate frequency outputs comprises using a programmable switch matrix to selectively provide intermediate frequency outputs from any of the downconverters to at least one demodulator in any of multiple different groups of demodulators.
14. The method of claim 10, wherein distributing the intermediate frequency outputs comprises using frequency dividers for each downconverter, and the frequency dividers providing the output of a downconverter to each of multiple sets of demodulators.
15. The method of claim 10, wherein the multiple polarizations include right-hand circular polarization and left-hand circular polarization.
16. The method of claim 10, wherein the signals are received in the Q band over a bandwidth of at least 4 GHz.
17. The method of claim 10, further comprising receiving data for a plurality of different channel segments, wherein each of the multiple intermediate frequency outputs of the downconverters includes modulated signals for multiple of the channel segments.
18. The method of claim 17, wherein the signals are received for each of the multiple polarizations in multiple channel segments each having bandwidth between 250 MHz and 2 GHz.

